

SAMPLE PAPER – 1 OF TCS NQT EXAM

NUMERICAL ABILITY

1. The LCM of two numbers is 2310 and their HCF is 30. If one of the numbers is 210, find the other.

- (A) 330
- (B) 310
- (C) 390
- (D) 270

Answer: (A) 330

2. How many two-digit numbers are divisible by both 3 and 7?

- (A) 4
- (B) 5
- (C) 6
- (D) 7

Answer: (B) 5

3. The price of sugar increases by 25%. By what percent should a family reduce its consumption so that expenditure remains unchanged?

- (A) 20%
- (B) 25%
- (C) 15%
- (D) 50%

Answer: (A) 20%

4. In an election between two candidates, one gets 60% of valid votes. If 15% votes are invalid and total votes are 10000, by how many votes does the winner win?

- (A) 2550
- (B) 1700
- (C) 2100
- (D) 5000

Answer: (A) 2550

5. A shopkeeper bought 200 eggs at ₹5 each. 20 eggs broke. He sold the rest at ₹6.50 each.

Find his profit%.

- (A) 17%
- (B) 14.5%
- (C) 16.25%
- (D) 18%

Answer: (A) 17%

6. An article is sold at 20% profit. If both cost price and selling price are decreased by ₱50, the profit% becomes 50%. Find the original cost price.

- (A) ₱200
- (B) ₱250
- (C) ₱150
- (D) ₱180

Answer: (B) ₱250

7. A trader marks his goods 40% above cost price and offers a 10% discount. Find the profit%.

- (A) 26%
- (B) 30%
- (C) 24%
- (D) 28%

Answer: (A) 26%

8. Two trains 140m and 160m long run at speeds of 60 km/h and 40 km/h in opposite directions. Time to cross each other?

- (A) 10.08 sec
- (B) 10.8 sec
- (C) 12 sec
- (D) 9.6 sec

Answer: (B) 10.8 sec

9. A man walks at 5 km/h and reaches 10 minutes late. If he walks at 6 km/h, he reaches 5 minutes early. Find the distance.

- (A) 7.5 km

- (B) 10 km
- (C) 12.5 km
- (D) 6 km

Answer: (A) 7.5 km

10. A and B together can do work in 12 days. B and C together in 15 days. A and C together in 20 days. How long will A alone take?

- (A) 20 days
- (B) 24 days
- (C) 30 days
- (D) 16 days

Answer: (B) 24 days

11. Two pipes A and B fill a tank in 20 and 30 min. Pipe C empties it in 15 min. If all are opened, how long to fill the tank?

- (A) 60 min
- (B) 75 min
- (C) 90 min
- (D) 120 min

Answer: (D) 120 min

12. The ratio of A:B:C is 2:3:5. If their sum is 1000, how much more does C have than A?

- (A) 300
- (B) 350
- (C) 200
- (D) 250

Answer: (A) 300

13. The average of 8 numbers is 20. If each number is multiplied by 3, what is the new average?

- (A) 60
- (B) 63
- (C) 20
- (D) 80

Answer: (A) 60

14. A mixture has milk and water in ratio 5:3. 16 litres of the mixture is removed and replaced with water. Ratio becomes 3:5. Find original quantity.

(A) 40 litres

(B) 48 litres

(C) 32 litres

(D) 56 litres

Answer: (A) 40 litres

15. The area of a circle is equal to the area of a square of side 14 cm. What is the circumference of the circle? ($\pi = 22/7$)

(A) 56 cm

(B) 52.5 cm

(C) 44 cm

(D) 88 cm

Answer: (C) 44 cm

16. In a right triangle, one leg is 8 cm and hypotenuse is 17 cm. Find the area of the triangle.

(A) 60 sq cm

(B) 68 sq cm

(C) 52 sq cm

(D) 75 sq cm

Answer: (A) 60 sq cm

17. In what ratio must rice at ₹40/kg be mixed with rice at ₹60/kg so that the mixture costs ₹50/kg?

(A) 1:1

(B) 2:3

(C) 3:2

(D) 1:2

Answer: (A) 1:1

18. How many 3-digit numbers can be formed using digits 1, 2, 3, 4, 5 without repetition?

(A) 60

(B) 120

(C) 20

(D) 80

Answer: (A) 60

19. A bag has 5 red, 4 blue and 5 green balls. If 2 balls are drawn at random, what is the probability that both are red?

(A) $\frac{10}{66}$

(B) $\frac{5}{33}$

(C) $\frac{2}{11}$

(D) $\frac{1}{6}$

Answer: (A) $\frac{10}{66}$

20. The sum of ages of father and son is 56 years. After 4 years, the father will be 3 times as old as the son. Find the current age of the son.

(A) 10 years

(B) 11 years

(C) 12 years

(D) 8 years

Answer: (C) 12 years

VERBAL ABILITY

Reading Comprehension — 5 Questions (1 Passage)

PASSAGE: Technology has transformed the way humans interact with one another. Social media platforms allow individuals to maintain connections across distances that were once prohibitive. However, critics argue that these platforms create shallow relationships, replacing meaningful face-to-face interactions with fleeting digital exchanges. Studies have shown that heavy social media use correlates with increased feelings of loneliness and reduced life satisfaction. Proponents counter that for geographically separated families and friends, digital communication provides a vital lifeline. The debate continues, as researchers examine how digital habits shape psychological well-being.

21. What is the primary concern raised by critics of social media in the passage?

- (A) Lack of internet access for all
- (B) Shallow digital relationships replacing deep ones
- (C) Technology addiction among teenagers
- (D) Decreased economic productivity

Answer: (B) Shallow digital relationships replacing deep ones

22. Which group is said to benefit most from digital communication?

- (A) Online gamers
- (B) Remote employees
- (C) Geographically separated families and friends
- (D) Social activists

Answer: (C) Geographically separated families and friends

23. What does the word 'prohibitive' most closely mean in the passage?

- (A) Encouraging
- (B) Excessive or restrictive
- (C) Affordable
- (D) Common

Answer: (B) Excessive or restrictive

24. What is the author's stance in the passage?

- (A) Strongly pro-social media
- (B) Strongly anti-social media
- (C) Balanced presentation of both views
- (D) Indifferent to the debate

Answer: (C) Balanced presentation of both views

25. What aspect does the passage say researchers are examining?

- (A) Economic impact of social media
- (B) Effect of digital habits on psychological well-being
- (C) Cybersecurity threats
- (D) Online privacy laws

Answer: (B) Effect of digital habits on psychological well-being

26. Despite being warned repeatedly, he _____ his reckless behaviour.

(A) persevered in

(B) persisted in

(C) insisted on

(D) continued on

Answer: (B) persisted in

27. The committee has _____ its decision to postpone the annual conference.

(A) revoked

(B) ratified

(C) deliberated

(D) escalated

Answer: (A) revoked

28. She was _____ by the company's sudden decision to terminate her contract without prior notice.

(A) overjoyed

(B) indifferent

(C) flabbergasted

(D) motivated

Answer: (C) flabbergasted

29. The scientist's research was so _____ that it opened entirely new fields of inquiry.

(A) mundane

(B) redundant

(C) groundbreaking

(D) trivial

Answer: (C) groundbreaking

30. Although he was _____ about the new policy, he chose to comply with it.

(A) enthusiastic

(B) ambivalent

(C) oblivious

(D) appreciative

Answer: (B) ambivalent

31. Identify the error: 'The news about the floods are disturbing.'

- (A) The news
- (B) about the floods
- (C) are disturbing
- (D) No error

Answer: (C) are disturbing

32. Choose the correct sentence:

- (A) He don't know the answer.
- (B) They was playing outside.
- (C) She has submitted her report.
- (D) We is going to the market.

Answer: (C) She has submitted her report.

33. Spot the error: 'Neither of the two boys were present at the meeting.'

- (A) Neither of
- (B) the two boys
- (C) were present
- (D) at the meeting

Answer: (C) were present

34. Which sentence is grammatically correct?

- (A) He is one of the student who work hard.
- (B) He is one of the students who works hard.
- (C) He is one of the students who work hard.
- (D) He is one of student who works hard.

Answer: (C) He is one of the students who work hard.

38. SYNONYM OF PERFIDIOUS:

- (A) Loyal
- (B) Courageous
- (C) Treacherous
- (D) Honest

Answer: (C) Treacherous

39. ANTONYM OF EPHEMERAL:

- (A) Transient
- (B) Brief
- (C) Temporary
- (D) Permanent

Answer: (D) Permanent

40. Arrange: P: He decided to take a different route. Q: The usual road was blocked due to construction. R: He finally reached his office on time. S: There was heavy traffic on the alternate route too.

- (A) QPSR
- (B) PQRS
- (C) SPQR
- (D) RQPS

Answer: (A) QPSR

41. Arrange: P: Reading improves vocabulary. Q: It also develops critical thinking. R: Books are a valuable source of knowledge. S: That is why experts recommend reading daily.

- (A) RPQS
- (B) QPRS
- (C) RSPQ
- (D) PQRS

Answer: (A) RPQS

42. Which sentence best fills the blank? 'Global warming poses serious risks to coastal cities. _____ Scientists warn that urgent action is needed.'

- (A) Tourism has declined in these areas.
- (B) Rising sea levels threaten millions of residents.
- (C) Clean energy is becoming more affordable.
- (D) Arctic exploration has expanded significantly.

Answer: (B) Rising sea levels threaten millions of residents.

43. Choose the correctly spelled word:

- (A) Accomodation

(B) Accommodation

(C) Accomodation

(D) Adcomodation

Answer: (B) Accommodation

44. Choose the word that means 'the quality of being honest and having strong moral principles':

(A) Arrogance

(B) Integrity

(C) Lethargy

(D) Frivolity

Answer: (B) Integrity

45. Fill the blank: She has been working at this company _____ 2015.

(A) for

(B) from

(C) since

(D) during

Answer: (C) since

REASONING ABILITY

46. Find the missing number: 3, 6, 11, 18, 27, ?

(A) 36

(B) 38

(C) 37

(D) 40

Answer: (B) 38

47. Find the missing term: 2, 3, 5, 8, 13, 21, ?

(A) 30

(B) 34

(C) 32

(D) 28

Answer: (B) 34

48. What comes next? 144, 121, 100, 81, 64, ?

(A) 49

(B) 56

(C) 50

(D) 48

Answer: (A) 49

49. Find the next number: 1, 8, 27, 64, 125, ?

(A) 196

(B) 216

(C) 200

(D) 210

Answer: (B) 216

50. Find the next: AZ, BY, CX, DW, ?

(A) EV

(B) FU

(C) EW

(D) GV

Answer: (A) EV

51. Find the next: A, E, I, M, Q, ?

(A) T

(B) U

(C) V

(D) W

Answer: (B) U

52. Identify the next: ABD, CEG, FIL, ?

(A) IMP

(B) INP

(C) KMP

(D) IMQ

Answer: (A) IMP

53. If FRIEND is coded as HUMJTK, how is CANDLE coded?

(A) EDRIRL

(B) EDRIRN

(C) ECPENG

(D) EDRIJL

Answer: (A) EDRIRL

54. In a code, 'go to school' = 3 5 7; 'school is fun' = 7 9 1; 'fun go always' = 1 5 2. What is the code for 'to'?

(A) 9

(B) 3

(C) 5

(D) 2

Answer: (B) 3

55. If in a code TIGER = 84... What does RIVER give?

(A) REVVI

(B) REVVW

(C) IREEV

(D) RIVVI

Answer: (A) REVVW

56. Pointing to a photograph, Raj says, 'She is the daughter of the wife of my father's only son.' How is the girl in the photo related to Raj?

(A) Sister

(B) Daughter

(C) Niece

(D) Wife

Answer: (B) Daughter

57. A is B's sister. C is B's mother. D is C's father. E is D's mother. How is A related to D?

(A) Daughter

(B) Granddaughter

(C) Grandmother

(D) Great-granddaughter

Answer: (B) Granddaughter

58. Six people A, B, C, D, E, F sit in a row. A is 3rd from the left. B is next to A on the right. C is at the extreme right. D is between E and F. E is 2nd from left. Who is 4th from the right?

(A) A

(B) B

(C) D

(D) F

Answer: (B) B

59. In a circular table of 6 people, P sits opposite Q. R is to the left of P. S is to the right of Q. T is between P and S (going clockwise). Who sits opposite R?

(A) S

(B) T

(C) Q

(D) E

Answer: (A) S

60. All mangoes are fruits. All fruits are nutritious. Conclusion: All mangoes are nutritious.

(A) Definitely True

(B) Definitely False

(C) Possibly True

(D) Cannot be determined

Answer: (A) Definitely True

61. Statements: Some cats are dogs. No dog is a horse. Conclusions: I. No cat is a horse. II. Some cats are not horses.

(A) Only I

(B) Only II

(C) Both I and II

(D) Neither I nor II

Answer: (B) Only II

62. All pens are books. Some books are boxes. Conclusions: I. Some pens are boxes. II.

Some boxes are pens.

- (A) Only I follows
- (B) Only II follows
- (C) Either I or II follows
- (D) Neither follows

Answer: (D) Neither follows

63. Ramesh walks 20m North, turns right and walks 50m, turns right again and walks 35m, then turns left and walks 15m. In which direction is he from the starting point?

- (A) East
- (B) West
- (C) North-East
- (D) South-East

Answer: (D) South-East

64. Book : Library :: Painting : ?

- (A) Artist
- (B) Canvas
- (C) Gallery
- (D) Museum

Answer: (C) Gallery

65. Find the odd one out: 8, 27, 64, 100, 125, 216

- (A) 64
- (B) 100
- (C) 216
- (D) 125

Answer: (B) 100

ADVANCED QUANTITATIVE ABILITY

66. The HCF of 2472, 1284 and N is 12. Their LCM is $2^3 \times 3^2 \times 5^2 \times 5^2 \times 103 \times 107$. What is N?

(A) 60

(B) 48

(C) 72

(D) 56

Answer: (A) 60

67. A milkman mixes milk costing ₱30/litre with water (cost ₱0) to make a mixture worth ₱24/litre. What is the ratio of milk to water?

(A) 4:1

(B) 5:1

(C) 3:1

(D) 2:1

Answer: (A) 4:1

68. A sphere of radius 6 cm is melted and recast into small spheres of radius 2 cm. How many small spheres are formed?

(A) 27

(B) 18

(C) 9

(D) 36

Answer: (A) 27

69. Two circles of radii 5 cm and 3 cm touch each other externally. What is the distance between their centres?

(A) 8 cm

(B) 2 cm

(C) 10 cm

(D) 6 cm

Answer: (A) 8 cm

70. Five years ago, the ratio of ages of A and B was 3:4. Five years hence it will be 4:5. What is the present age of B?

(A) 35

(B) 50

(C) 40

(D) 25

Answer: (A) 45 (Note: Image says A, but the text "35" aligns with answer A)

71. At what time between 4 and 5 o'clock will the minute and hour hands of a clock be perpendicular?

(A) 4:5.45

(B) 4:21.8

(C) 4:32.7

(D) 4:43.6

Answer: (B) 4:21.8

72. A card is drawn from a pack of 52 cards. What is the probability that it is a king or a red card?

(A) $\frac{28}{52}$

(B) $\frac{7}{13}$

(C) $\frac{28}{54}$

(D) $\frac{15}{26}$

Answer: (B) $\frac{7}{13}$

73. In how many ways can the letters of the word 'ARRANGE' be arranged?

(A) 1260

(B) 2520

(C) 840

(D) 5040

Answer: (A) 1260

74. The sum of first n terms of an AP is $\$3n^2 + 4n\$$. What is the common difference?

(A) 3

(B) 6

(C) 5

(D) 7

Answer: (B) 6

75. A man rows 36 km upstream in 9 hours and 48 km downstream in 8 hours. Find the

speed of the current.

- (A) 1 km/h
- (B) 1.5 km/h
- (C) 2 km/h
- (D) 0.5 km/h

Answer: (A) 1 km/h

ADVANCED REASONING ABILITY

76. Find the missing term: 2, 5, 10, 17, 26, 37, ?

- (A) 48
- (B) 50
- (C) 52
- (D) 54

Answer: (B) 50

77. Find the missing number: 6, 11, 21, 36, 56, ?

- (A) 81
- (B) 76
- (C) 91
- (D) 71

Answer: (A) 81

78. What replaces ?: 1, 2, 6, 24, 120, ?

- (A) 620
- (B) 720
- (C) 600
- (D) 480

Answer: (B) 720

79. In a code: STRONG = ROTNSG, MOBILE = IBOML E. How is GARDEN coded?

- (A) RAGNED
- (B) RAGDEN
- (C) RAGEND

(D) RAGNDE

Answer: (A) RAGNED

80. BEIM : ORUX :: CGKQ : ?

(A) PVZE

(B) PVZD

(C) QWZF

(D) PVZD

Answer: (A) PVZE

81. Decode: If CLOCK = DNQLT, what does WATCH decode to in the same pattern?

(A) XBVDI

(B) XBUFI

(C) YBVDI

(D) XBUFJ

Answer: (A) XBVDI

82. Eight people A-H sit around a circle facing centre... Who is between A and C (going clockwise)?

(A) G

(B) H

(C) E

(D) F

Answer: (A) G

83. Seven people P, Q, R, S, T, U, V sit in a row... Who is 4th from left?

(A) R

(B) T

(C) Q

(D) V

Answer: (B) T

84. If A&B means A is the father of B... What does P&Q+R@S mean?

(A) P is grandfather of S

(B) P is father of R's husband's sister

(C) R is sister of Q who is wife of P's son

(D) P is father of Q who is husband of R who is sister of S

Answer: (D)

85. Given: M+N means M is the mother of N... How is A related to D?

(A) Aunt

(B) Mother

(C) Sister

(D) Grandmother

Answer: (D) Grandmother

CODING

1. Find the Second Largest Element

Problem Statement: Given an array of N integers, your task is to find the second largest distinct element in the array. If there is no second largest distinct value, print -1.

Input Format:

- First line: Integer N — size of array
- Second line: N space-separated integers

Example 1:

Input: 6, 12 35 1 10 34 1

Output: 34

2. Count Pairs with Given Sum

Problem Statement: Given an array of N integers and a target value T, count the number of pairs (i, j) such that: $i < j$ and $arr[i] + arr[j] = T$.

Input Format:

- First line: Integer N
- Second line: Integer T
- Third line: N space-separated integers

Example:

Input: 6, 6, 1

Important Formulas for TCS NQT 2026

1. Number System: $\text{LCM} \times \text{HCF} = \text{Product of two numbers } (a \times b)$
2. Average: $\text{Sum of observations} / \text{Number of observations}$
3. Average Speed: $2xy / (x + y)$ (when distances are equal)
4. Percentage Change: $(\text{Change} / \text{Original Value}) \times 100$
5. Successive Percentage Change: $[a + b + (ab / 100)]\%$
6. Profit %: $(\text{Profit} / \text{CP}) \times 100$
7. Selling Price (SP): $\text{CP} \times (100 + P\%) / 100$ or $\text{CP} \times (100 - L\%) / 100$
8. Time, Speed & Distance: $\text{Speed} = \text{Distance} / \text{Time}$
9. Relative Speed (Opposite Directions): $S_1 + S_2$
10. Relative Speed (Same Direction): $|S_1 - S_2|$
11. Time & Work: If A takes x days, A's 1-day work $= 1/x$
12. Combined Work (A+B): $(xy) / (x+y)$ days
13. Arithmetic Progression (AP): $n\text{-th term} = a + (n-1)d$
14. Sum of n terms of AP: $(n/2)[2a + (n-1)d]$
15. Circle: $\text{Area} = \pi r^2$, $\text{Circumference} = 2\pi r$
16. Probability: $P(E) = \text{Number of favorable outcomes} / \text{Total outcomes}$
17. Combination: $nCr = n! / [r! (n-r)!]$
18. Clock Angle: $|30h - (11/2)m|$ where h = hours, m = minutes